CSE 4849
Human Computer Interaction (HCI)
Lecture 5: Conceptualizing Interaction

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Conceptualizing Interaction

- Imagine you have been asked to design an application to let people organize, store, and retrieve their email in a fast, efficient and enjoyable way.

- What would you do? How would you start?

Understanding the problem space

- Why, what and how?
  - What do you want to create?
  - What are your assumptions?
  - What are your claims?
  - Will it achieve what you hope it will? If so, how?

Understanding the problem space…

A Hypothetical Scenario of Early Design Highlighting the Assumptions and Claims (*italicized*) Made by Different Members of a Design Team

A large software company has decided it needs to develop an upgrade of its web browser for smartphones because its marketing team has discovered that many of the company's customers have switched over to using another mobile browser. The marketing people *assume* something is wrong with their browser and that their rivals have a better product. But they don't know what the problem is with theirs. The design team put in charge of this project *assume* they need to improve the usability of a number of the browser's functions. They *claim* that this will win back users by making features of the interface simpler, more attractive, and more flexible to use.

The user experience researchers on the design team conduct an initial user study investigating how people use the company's web browser on a variety of smartphones. They also look at other mobile web browsers on the market and compare their functionality and usability. They observe and talk to many different users. They discover several things about the usability of their web browser, some of which they were not expecting. One revelation is that many of their customers have never actually used the bookmarking tool. They present their findings to the rest of the team and have a long discussion about why each of them thinks it is not being used. One member *claims* that the web browser's function for organizing bookmarks is fiddly and error-prone and *assumes* this is the reason why many users do not use it. Another member backs her up, saying how awkward it is to use this method when wanting to move bookmarks between folders. One of the user experience architects agrees, noting how several of the users he talked to mentioned how difficult and time-consuming they found it when trying to move bookmarks between folders and how they often ended up accidentally putting them into the wrong folders.

A software engineer reflects on what has been said, and makes the *claim* that the bookmark function is no longer needed since he *assumes* that most people do what he does, which is to revisit a website by flicking through their history list of previously visited pages. Another member of the team disagrees with him, *claiming* that many users do not like to leave a trail of the sites they have visited and would prefer to be able to save only sites they think they might want to revisit. The bookmark function provides them with this option. Another option discussed is whether to include most-frequently visited sites as thumbnail images or as tabs. The software engineer agrees that providing all options could be a solution but worries how this might clutter the small screen interface.

After much discussion on the pros and cons of bookmarking versus history lists, the team decides to investigate further how to support effectively the saving, ordering, and retrieving of websites using a mobile web browser. All agree that the format of the existing web browser's structure is too rigid and that one of their priorities is to see how they can create a simpler way of revisiting websites on the smartphone.

Understanding the problem space…

- Based on this analysis of the previous scenario, a set of assumptions about the user needs for supporting this activity more effectively were then made. These included:
  - *If the bookmarking function was improved users would find it more useful and use it more to organize their web addresses.*
  - *Users need a flexible way of organizing web addresses they want to keep for further reference or for sending on to other people.*

A framework for analysing the problem space

- Are there problems with an existing product or user experience? If so, what are they?
- Why do you think there are problems?
- How do you think your proposed design ideas might overcome these?
- When designing for a new user experience how will the proposed design extend or change current ways of doing things?
- How will it support people in their activities?
- Will it really help them?

What is assumption and claim?

- Assumption:
  - Taking something for granted when it needs further investigation.
  - E.g. People will want to watch TV while driving
- Claim:
  - Stating something to be true when it still open to question.
  - E.g. a multimodal style of interaction for controlling GPS, one that involves speaking while driving, is safe.

An example

- What do you think were the main assumptions made by developers of online photo sharing and management applications, like Flickr?

Assumptions and claims of Flickr

- Assumptions
  - Able to capitalize on the hugely successful phenomenon of blogging
  - Just as people like to blog so will they want to share with the rest of the world their photo collections and get comments back
  - People like to share their photos with the rest of the world
- A claim
  - From Flickr’s website (2005): “is almost certainly the best online photo management and sharing application in the world”

Benefits of conceptualizing

- Orientation
  - enables design teams to ask specific questions about how the conceptual model will be understood
- Open-minded
  - prevents design teams from becoming narrowly focused early on
- Common ground
  - allows design teams to establish a set of commonly agreed terms

From problem space to design space

- Having a good understanding of the problem space can help inform the design space
  - e.g., what kind of interface, behavior, functionality to provide
- But before deciding upon these it is important to develop a conceptual model

Conceptual model

- A conceptual model is:

  “A high-level description of how a system is organized and operates.” (Johnson and Henderson, 2002, p. 26)

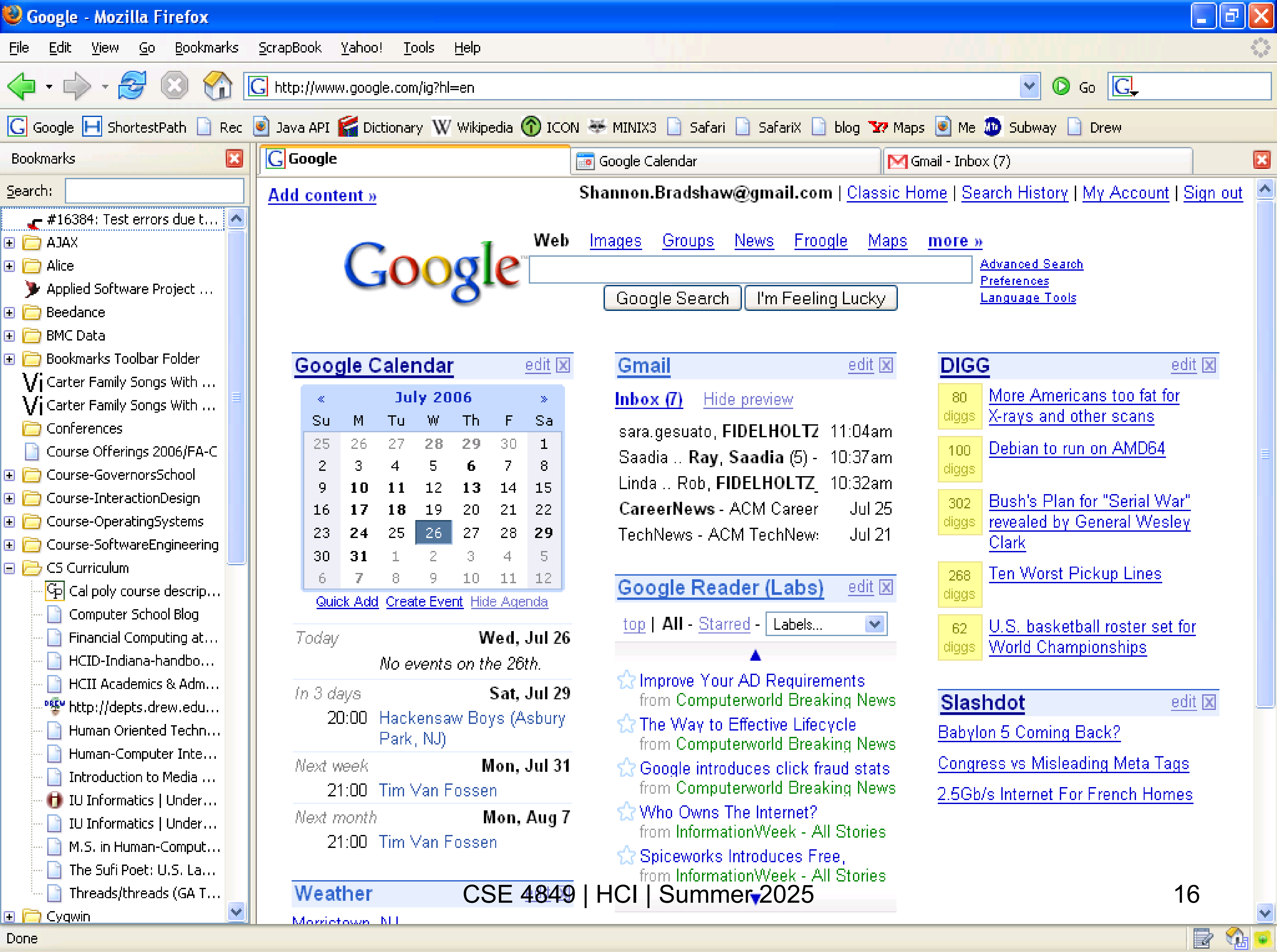
- Conceptual models enables:

  “designers to straighten out their thinking before they start laying out their widgets” (p. 28)

- Developing a conceptual model involves, envisioning the proposed product based on the users' needs and other requirements identified.
- Conceptual model based on objects
- Conceptual model based on activities
- Conceptual model based on interface metaphor.

Conceptual model

- Conceptual model is a strategy; a framework of general concepts and their interrelations.
- Johnson and Hnderson (2002) propose that, a conceptual model should comprise the following components:
  – **The major metaphor and analogies** to understand what a product is for and how to use it for an activity
  – **The concepts** that people are exposed to through the product, task–domain objects, their attributes, and operations (e.g. saving, revisiting, organizing)
  – The relationships and mapping between these concepts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Advanced Search
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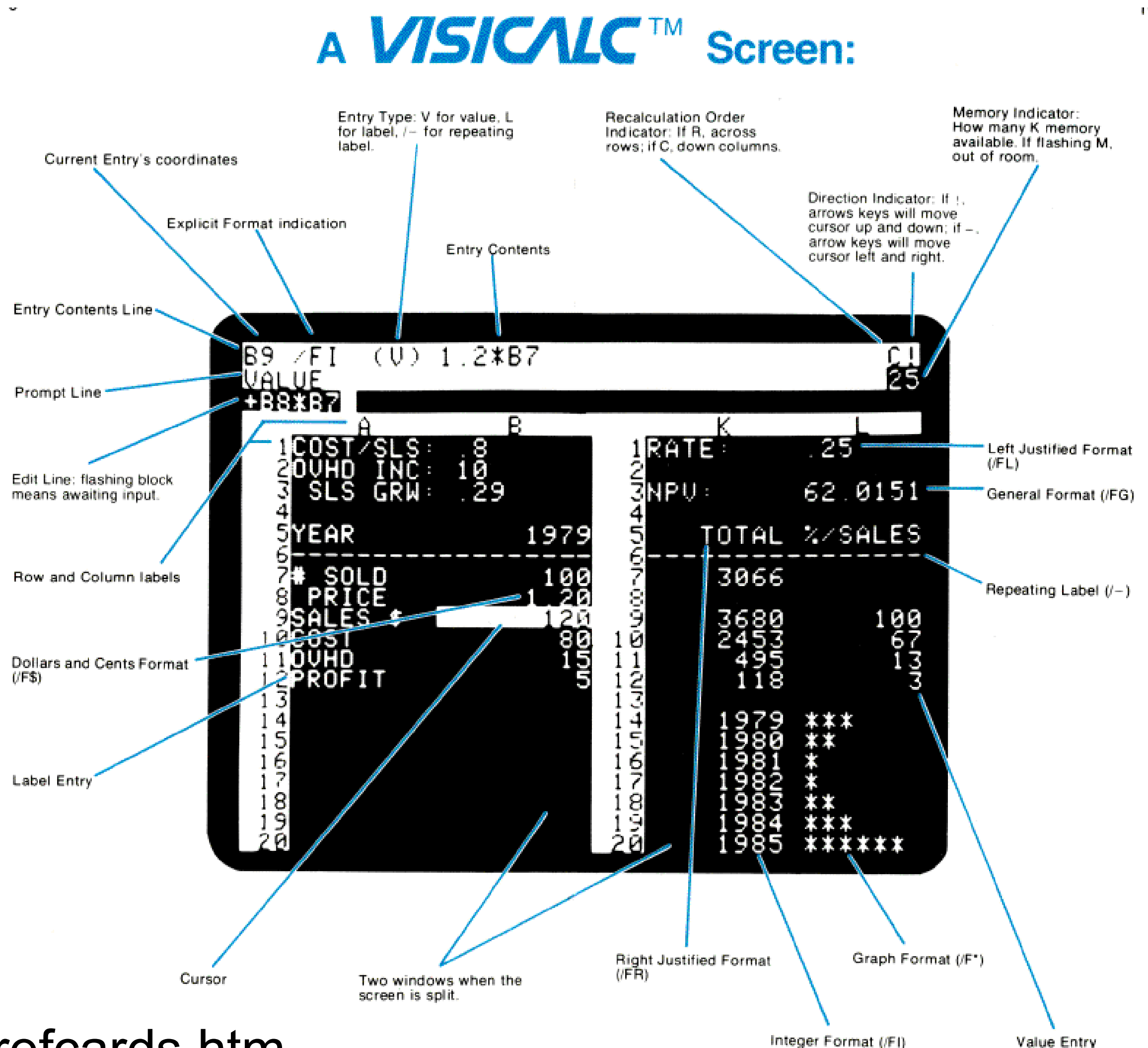
Google Search
I'm Feeling Lucky
Google Calendar edit
July 2006
Su M Tu W Th F Sa
Quick Add Create Event Hide Agenda
Today Wed, Jul 26
No events on the 26th.
In 3 days Sat, Jul 29
20:00 Hackensaw Boys (Asbury Park, NJ)
Next week Mon, Jul 31
21:00 Tim Van Fossen
Next month Mon, Aug 7
21:00 Tim Van Fossen
Weather
Morristown, NJ
Gmail edit
Inbox (7) Hide preview
sara.gesuato, FIDELHOLTZ 11:04am
Saadia .. Ray, Saadia (5) - 10:37am
Linda .. Rob, FIDELHOLTZ 10:32am
CareerNews - ACM Career Jul 25
TechNews - ACM TechNew: Jul 21
Google Reader (Labs) edit
top | All - Starred - Labels...
Improve Your AD Requirements
from Computerworld Breaking News
The Way to Effective Lifecycle
from Computerworld Breaking News
Google introduces click fraud stats
from Computerworld Breaking News
Who Owns The Internet?
from InformationWeek - All Stories
Spiceworks Introduces Free,
from InformationWeek - All Stories
DIGG edit
80 diggs More Americans too fat for X-rays and other scans
100 diggs Debian to run on AMD64
302 diggs Bush's Plan for "Serial War" revealed by General Wesley Clark
268 diggs Ten Worst Pickup Lines
62 diggs U.S. basketball roster set for World Championships
Slashdot edit
Babylon 5 Coming Back?
Congress vs Misleading Meta Tags
2.5Gb/s Internet For French Homes
Done

What is and why need a conceptual model?

- A conceptual model is not a description of the user interface but a structure outlining the concepts and the relationships between them
- Why not start with the nuts and bolts of design?
  - Architects and interior designers would not think about which colour curtains to have before deciding where the windows will be placed in a new building
  - Enables “designers to straighten out their thinking before they start laying out their widgets” (p. 28)
  - Provides a working strategy and a framework of general concepts and their interrelations
  - Mode of interaction vs. Style of interaction
  - Interface metaphor

A classic conceptual model: the spreadsheet

- **VisiCalc (Bricklin and Frankston)**
- Analogous to ledger sheet
- Interactive and computational
- Easy to understand
- Greatly extending what accountants and others could do



www.bricklin.com/history/refcards.htm

Why was it so good?

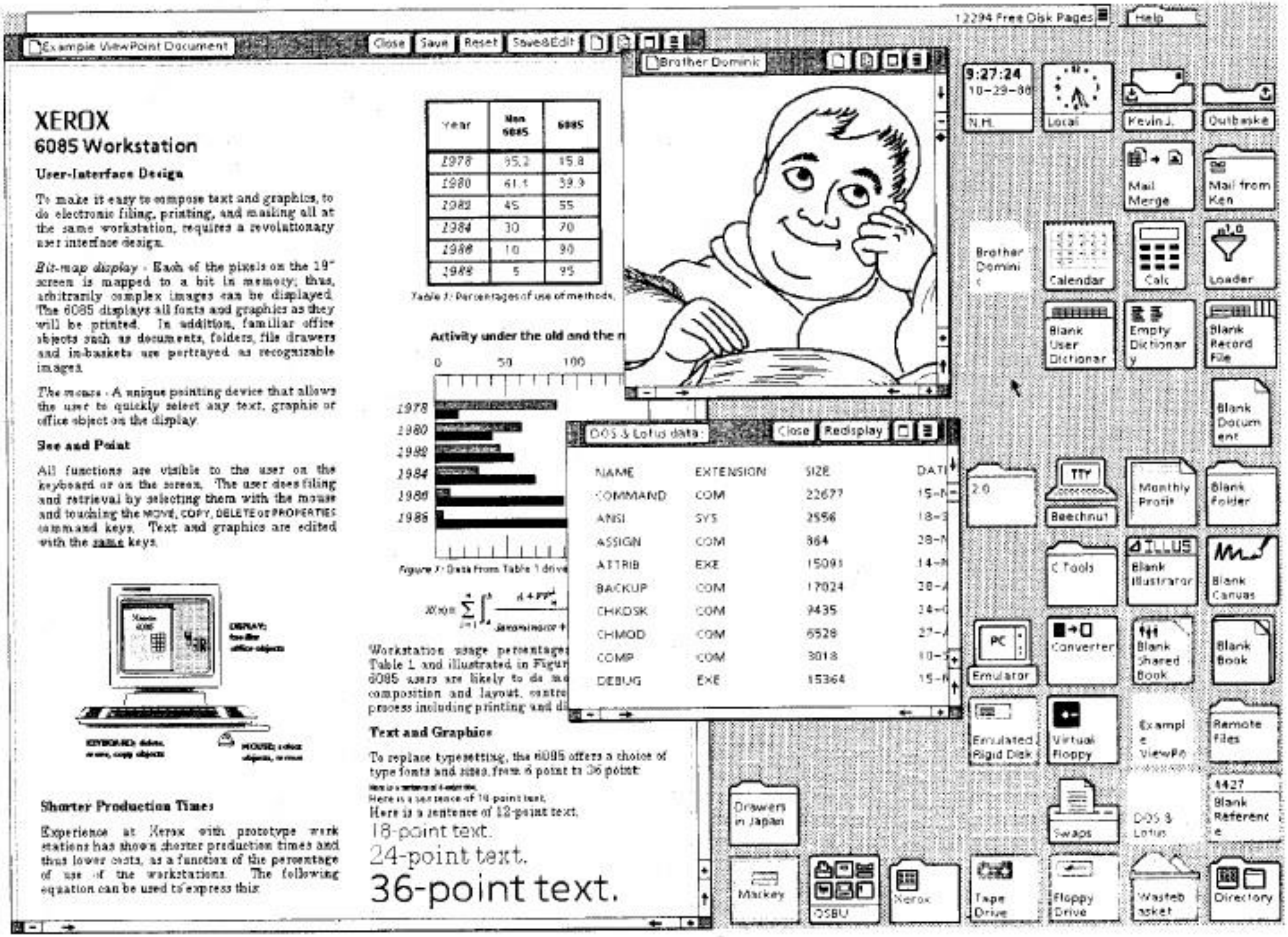
- It was simple, clear, and obvious to the users how to use the application and what it could do
- “it is just a tool to allow others to work out their ideas and reduce the tedium of repeating the same calculations.”
- Capitalized on user’s familiarity with ledger sheets
- Got the computer to perform a range of different calculations and recalculations in response to user input

Another classic

- 8010 Star office system targeted at workers not interested in computing *per se*
- Spent several person-years at beginning working out the conceptual model
- Simplified the electronic world, making it seem more familiar, less alien, and easier to learn

Johnson et al (1989)

The Star interface



Interface metaphors

- Interface designed inspired from know physical entity but also has its own properties.
  - E.g. desktop metaphor, web portals, mobile interface
- Metaphors are commonly used to explain something that is unfamiliar or hard to grasp by way of comparison with something that is familiar and easy to grasp.
- In general a model requires more learning
- A model requires more learning without metaphors to which users can anchor it to their previous experience.
- Metaphors are used in interaction design
- Interface metaphors are often composites

Conceptual model based on activities - Interaction types

- Instructing
  - Users issues instructions to a system
  - Typing in commands using keyboard and function keys, selecting options via menus, pressing buttons etc.
- Conversing
  - interacting with the system as if having a conversation via dialog
  - Users speak to the system or type in questions to which the system replies via text or speech output.
- Manipulating
  - interacting with objects in a virtual or physical space by manipulating them
  - Opening, holding, closing, placing.
- Exploring
  - Moving through a virtual environment or a physical space
  - Virtual reality systems and 3D world.

Instructing

- Where users instruct a system by telling it what to do
  - e.g., tell the time, print a file, find a photo
- Very common interaction type underlying a range of devices and systems
- A main benefit of instructing is to support quick and efficient interaction
  - good for repetitive kinds of actions performed on multiple objects

Vending machines

Describe the conceptual model underlying the two vending machines

Which is easiest to use?

Conversing

- Like having a conversation with another human
- Differs from instructing in that it more like two-way communication, with the system acting like a partner rather than a machine that obeys orders
- Ranges from simple voice recognition menu-driven systems to more complex ‘natural language’ dialogues
- Examples include advice-giving systems and help systems

Pros and cons of conversational model

- Allows users, especially novices and technophobes, to interact with the system in a way that is familiar
  - makes them feel comfortable, at ease and less scared

- Misunderstandings can arise when the system does not know how to parse what the user says
  - e.g. child types into a search engine, that uses natural language the question:

  "How many legs does a centipede have?" and the system responds:

Manipulating

- Exploit's users' knowledge of how they move and manipulate in the physical world

- Virtual objects can be manipulated by moving, selecting, opening, and closing them

Direct manipulation

- Shneiderman (1983) coined the term Direct Manipulation
- Came from his fascination with computer games at the time
- Proposes that digital objects be designed so they can be interacted with analogous to how physical objects are manipulated
- Assumes that direct manipulation interfaces enable users to feel that they are directly controlling the digital objects

Core principles of DM

- Continuous representation of objects and actions of interest
- Physical actions and button pressing instead of issuing commands with complex syntax
- Rapid reversible actions with immediate feedback on object of interest

Why are DM interfaces so enjoyable?

- Novices can learn the basic functionality quickly
- Experienced users can work extremely rapidly to carry out a wide range of tasks, even defining new functions
- Intermittent users can retain operational concepts over time
- Error messages rarely needed
- Users can immediately see if their actions are furthering their goals and if not do something else
- Users experience less anxiety
- Users gain confidence and mastery and feel in control

What are the disadvantages with DM?

- Some people take the metaphor of direct manipulation too literally
- Not all tasks can be described by objects and not all actions can be done directly
- Some tasks are better achieved through delegating rather than manipulating
  - e.g., spell checking
- Moving a mouse around the screen can be slower than pressing function keys to do same actions

Exploring

- Involves users moving through virtual or physical environments
- Examples include:
  - 3D desktop virtual worlds where people navigate using mouse around different parts to socialize (e.g., Second Life)
  - CAVEs where users navigate by moving whole body, arms, and head
  - physical context aware worlds, embedded with sensors, that present digital information to users at appropriate places and times

A virtual world

Summary points

- Need to have a good understanding of the problem space
  - specifying what it is you are doing, why, and how it will support users in the way intended
- A conceptual model is a high-level description of a product
  - what users can do with it and the concepts they need to understand how to interact with it
- Decisions about conceptual design should be made before commencing any physical design
- Interface metaphors are commonly used as part of a conceptual model
- Interaction types (e.g., conversing, instructing) provide a way of thinking about how best to support the activities users will be doing when using a product or service